

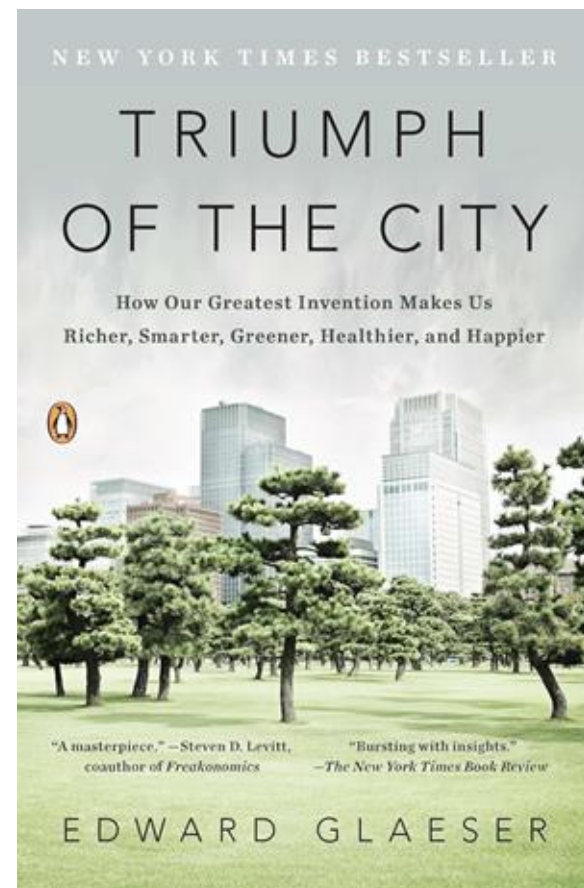
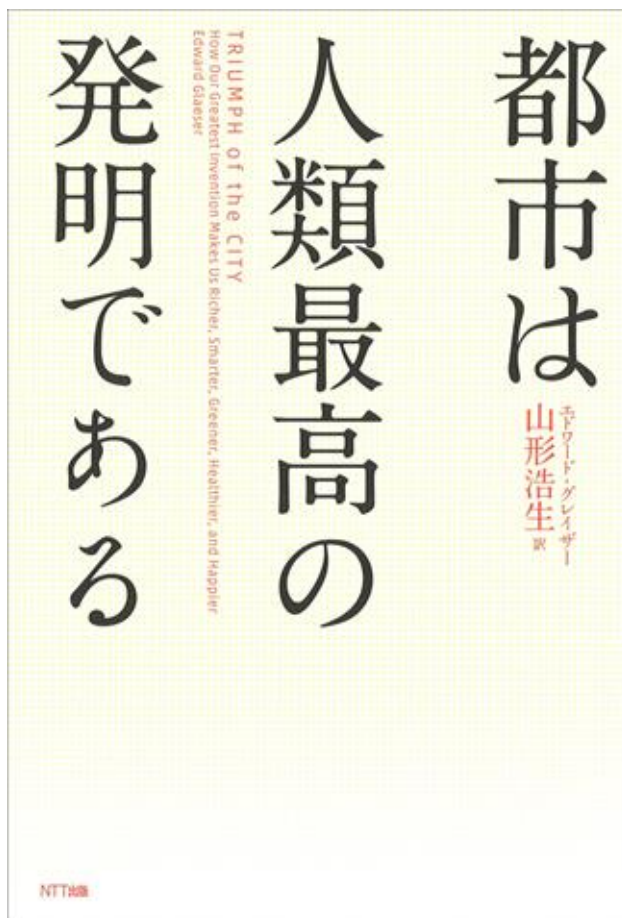
Open-Sparsification

**Spatial form for an era of population decline
and rising pandemic / disaster risk**

Kotaro Iwata

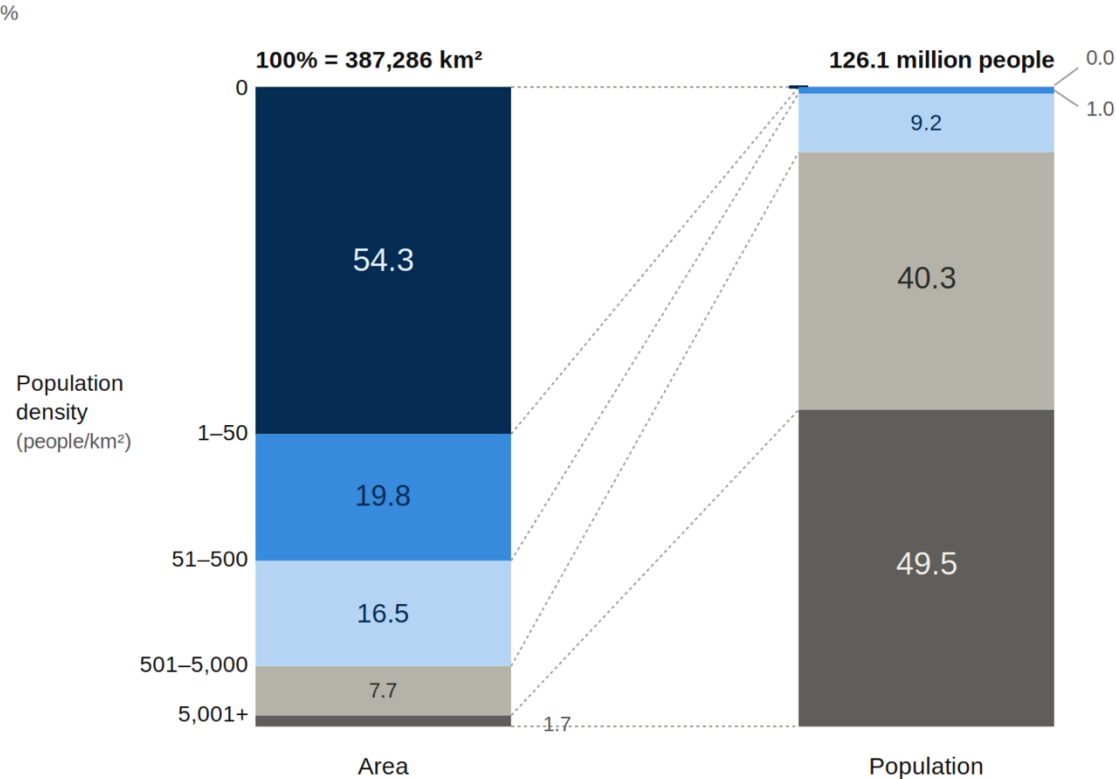
<https://github.com/memorist17/opensparsity>

Cities are wonderful



Half of the population lives in cities (Japan)

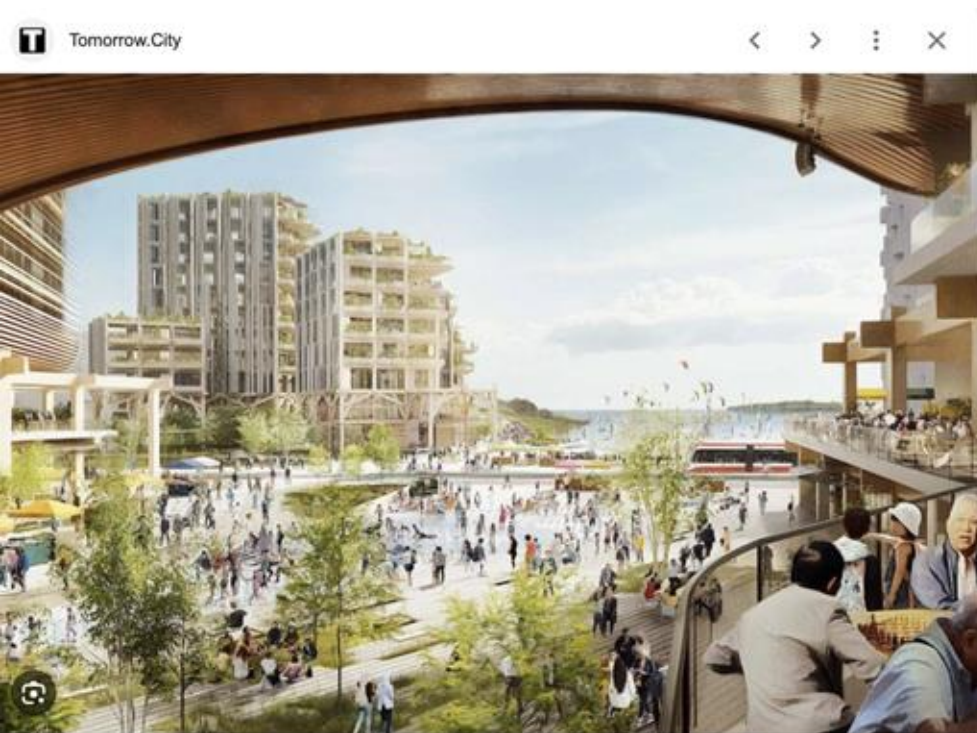
Area and population by population density



That works — if population grows and no disaster or pandemic hits
But that premise falls apart



not optimization



Sidewalk Toronto, the vision behind Google's failed city - Tomorrow.City - The biggest platform about urban innovation

移動 >



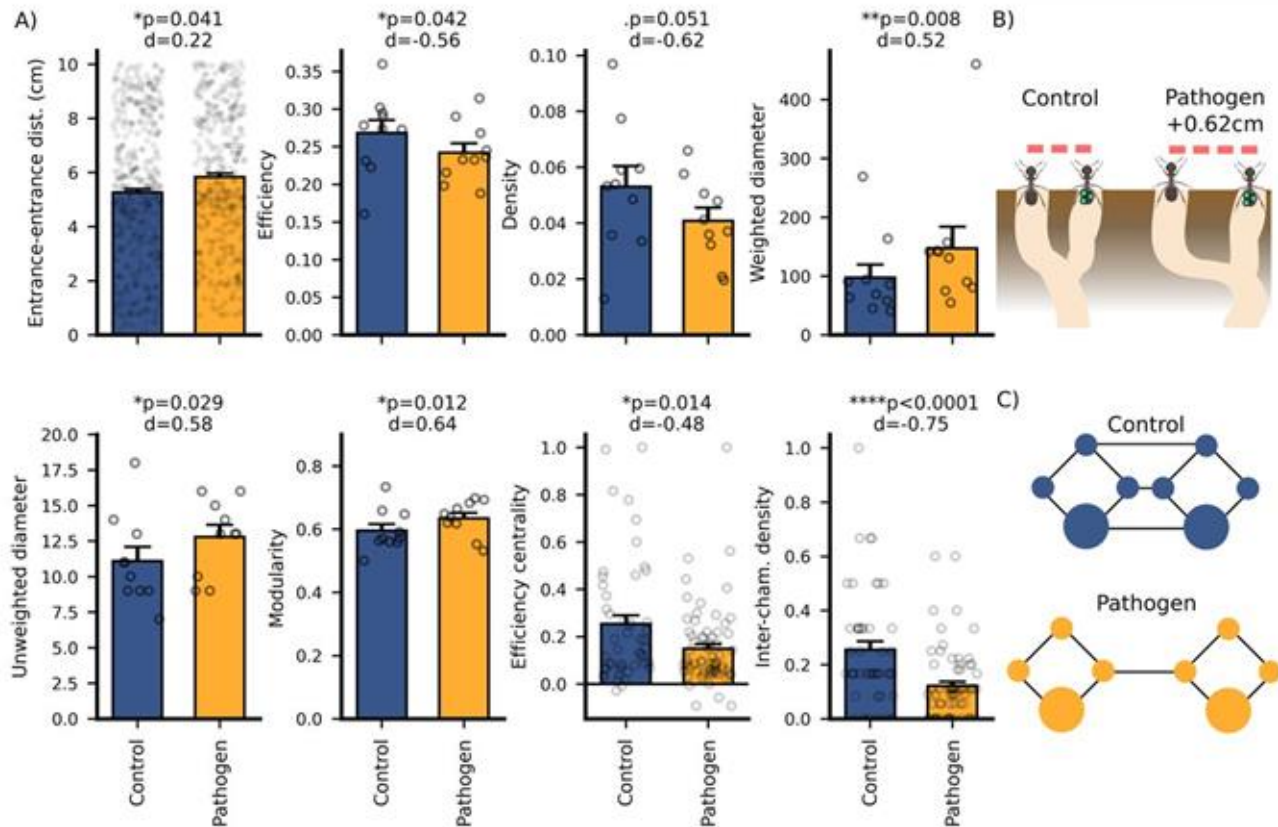
Masdar City: "The ecotopia" that never was | Green Prophet

移動 >

Insights from living systems

Ants restructure their nest against epidemics

Beyond lower density: centrality ↓, modularity ↑, and more



Humans, too,
should transform
settlement structure
as the environment changes

Research questions & scope: what I began this term

What kind of state is open sparsity? (indicators)

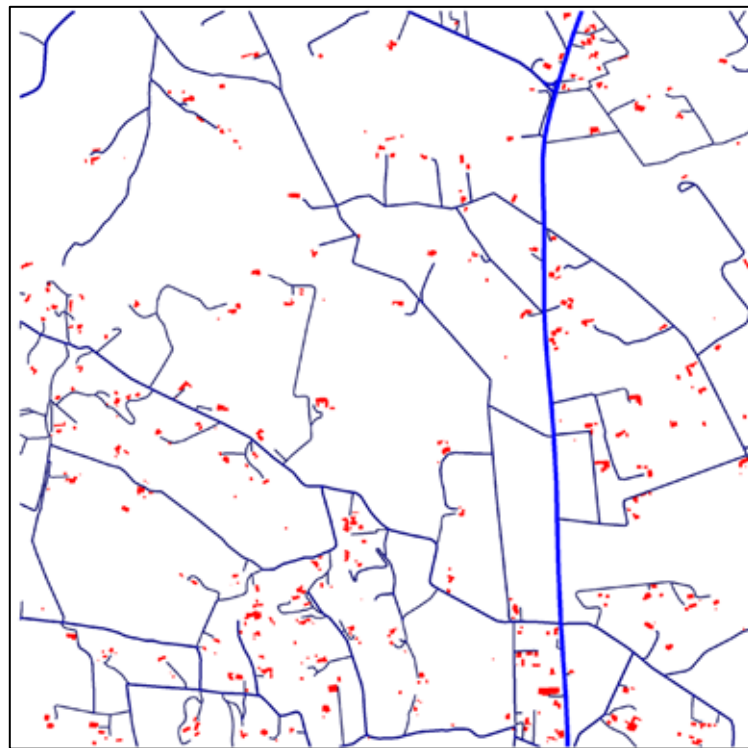
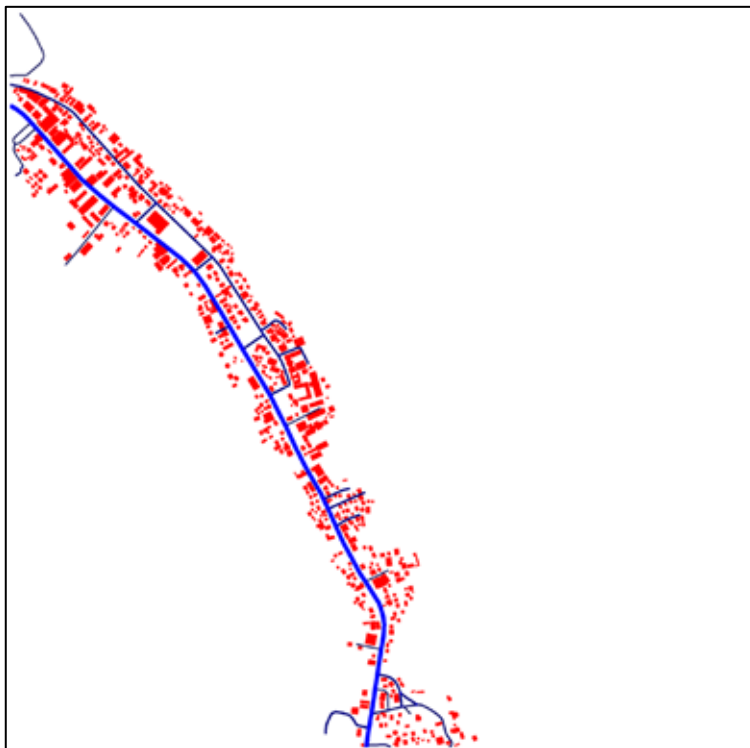
- What is open sparsity? (function, dynamics, form)
- Quantifying open-sparse form

How do we make space open-sparse? (rules)

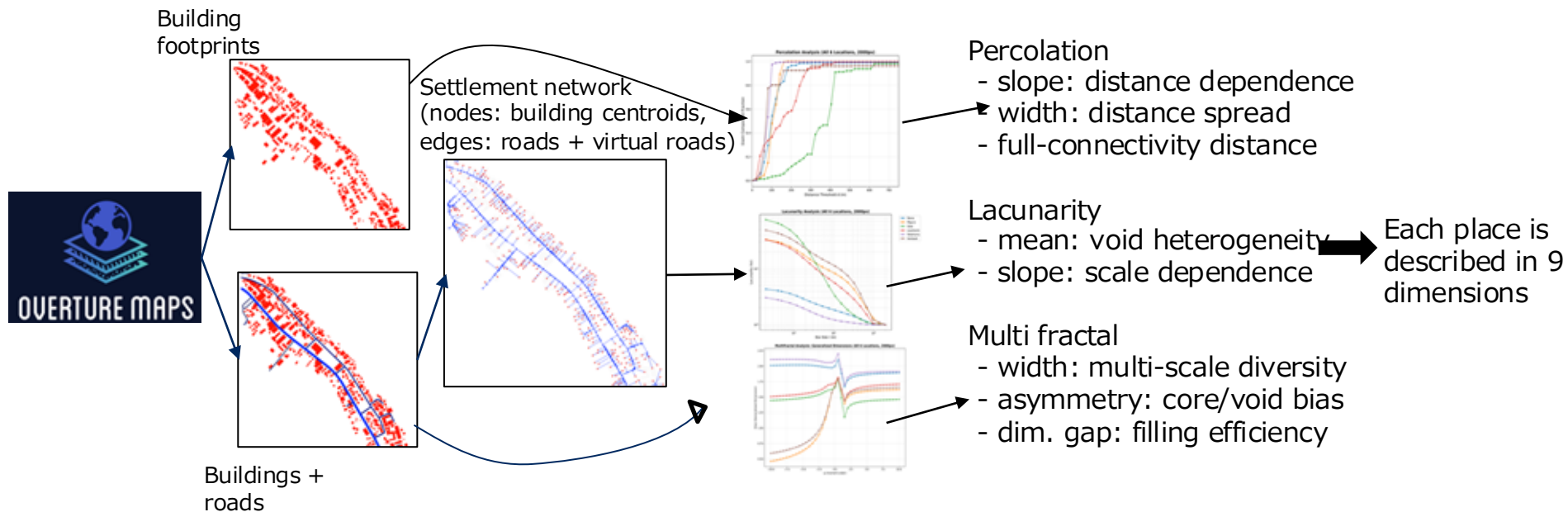
- Settlement variables that shape the rules
- Setting initial conditions
- Searching for concrete rules

Quantifying open-sparse form

Quantifying form $\doteq$
 same density, different structure — measured

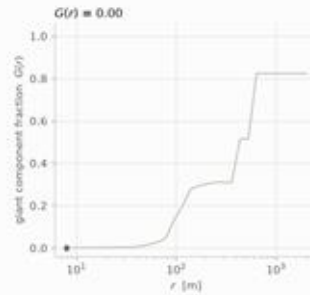


Measuring open-sparse form: building connectivity × void heterogeneity × multi-scale complexity



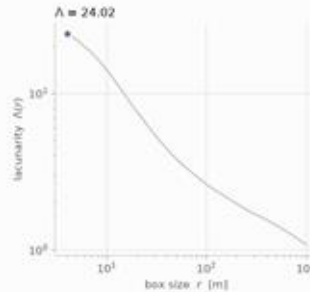
Percolation — two buildings are linked when the shortest path along the road network between them is $\leq r$

connection radius $r = 8$ m



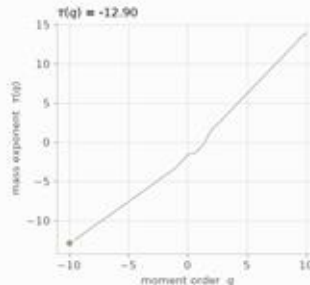
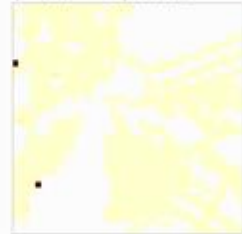
Lacunarity — slide a box of size r over the building raster: $\Lambda(r) = 1 + \sigma^2/\mu^2$ of the box masses

gliding box $r = 4$ m



Multifractal — weight each box by μ_i^q and watch which part of the settlement the exponent is listening to

$q = -10.0$ — emptiest boxes



Percolation

- slope: distance dependence
- width: distance spread
- full-connectivity distance

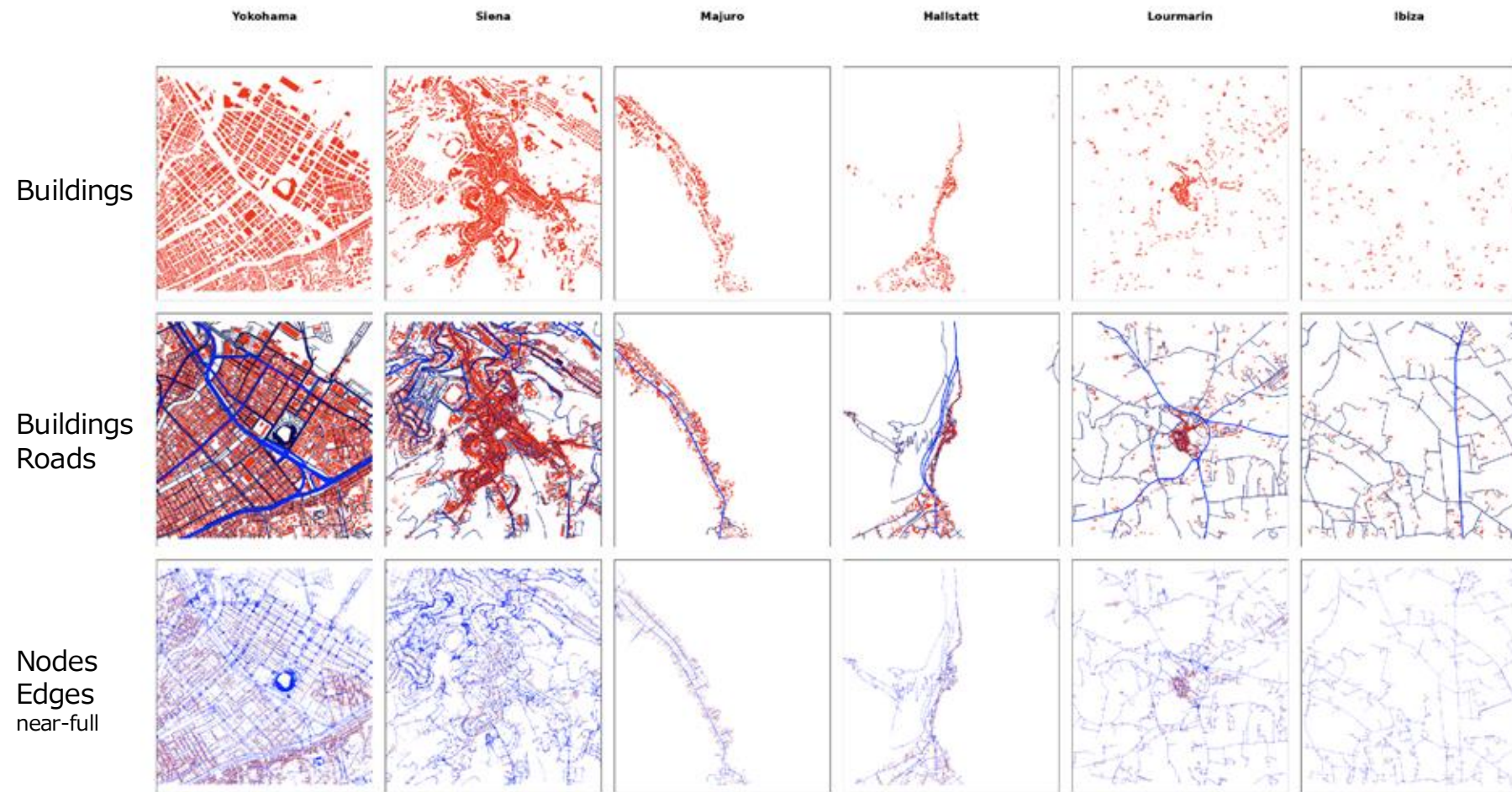
Lacunarity

- mean: void heterogeneity
- slope: scale dependence

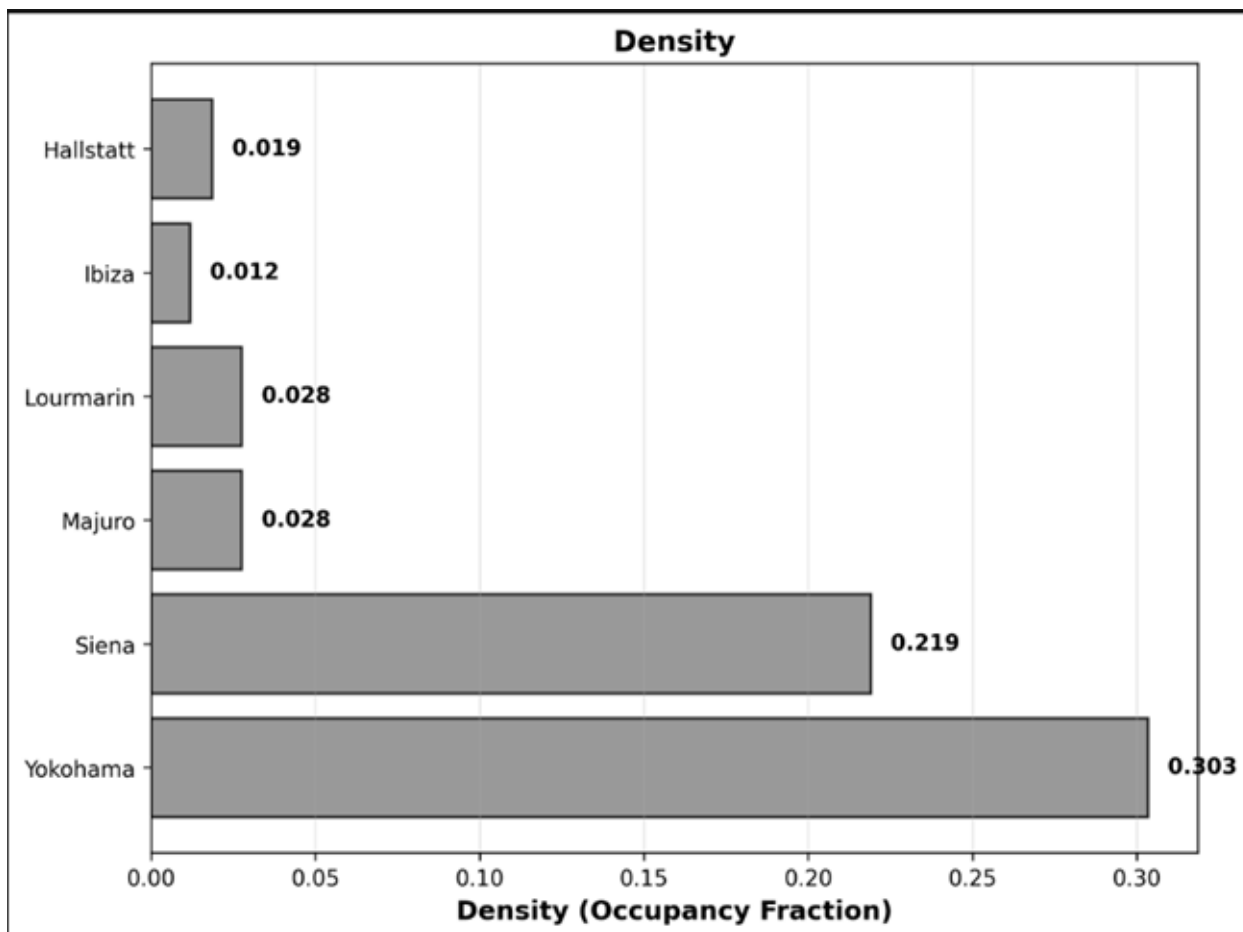
Multi fractal

- width: multi-scale diversity
- asymmetry: core/void bias
- dim. gap: filling efficiency

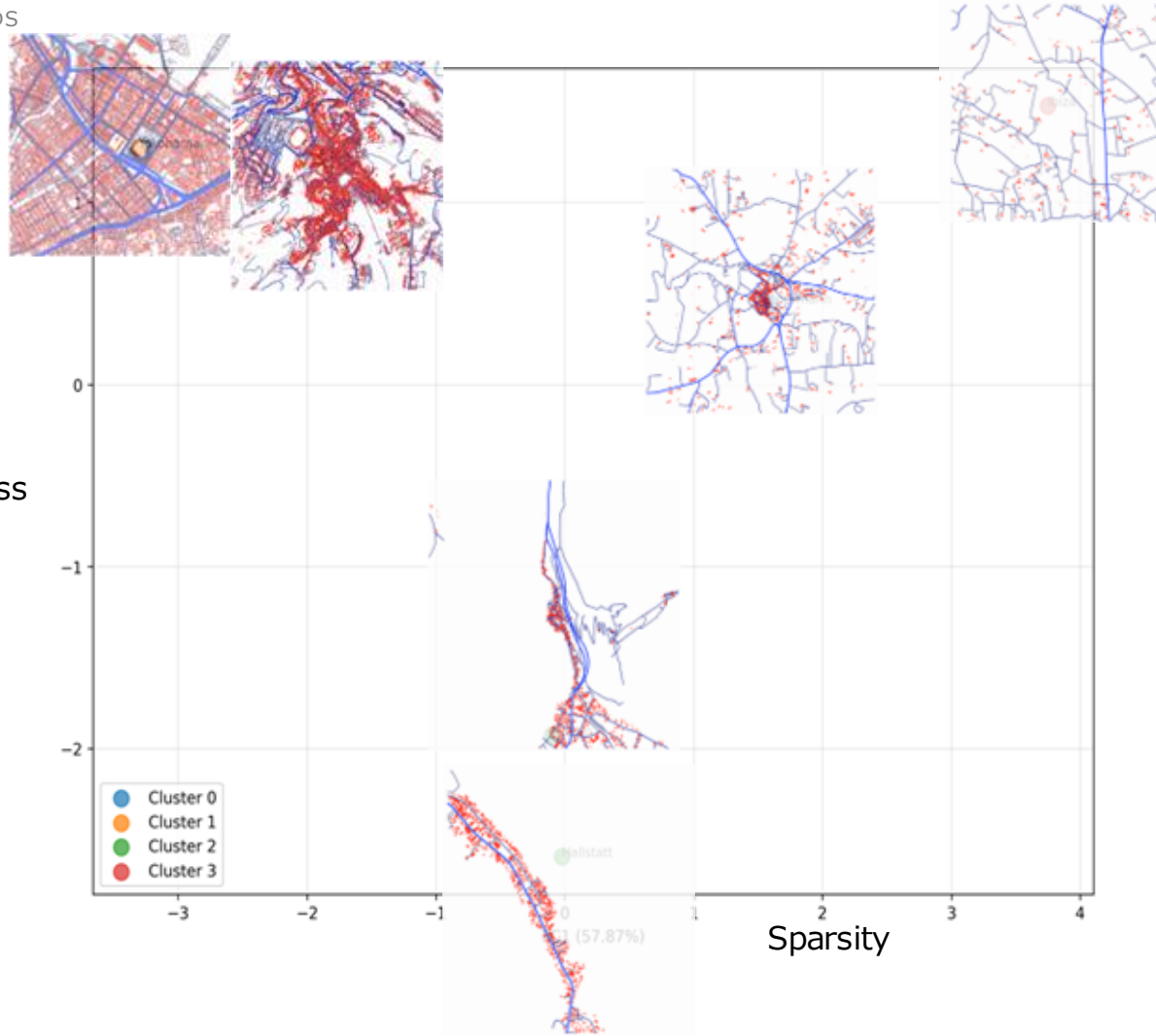
Computed and plotted the indicators for 6 characteristic sites



Density alone cannot separate Lourmarin and Majuro

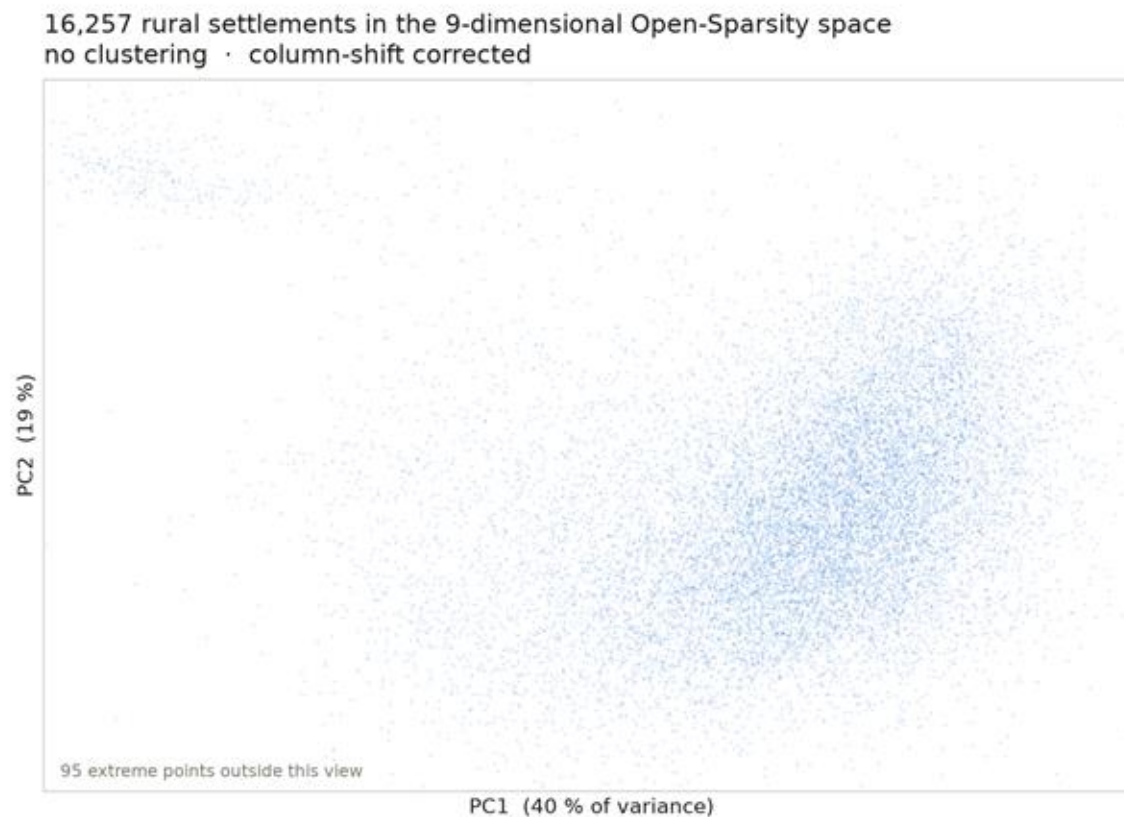


Openness



n = 16257 sites

Random sampling within a fixed density band



n = 16257

Random sampling within a fixed density band

The same 16,257 settlements, coloured by each indicator in turn — PCA layout, 0.5-99.5 % range · column-shift corrected

low → high



Next step

Next step

- Measure time series, not snapshots
- Generate settlements that do not yet exist
(settlement as it could be)
- Integrate the 3 layers of a settlement: function, dynamics, form
 - LLM agent

Takeaway

takeaway-1

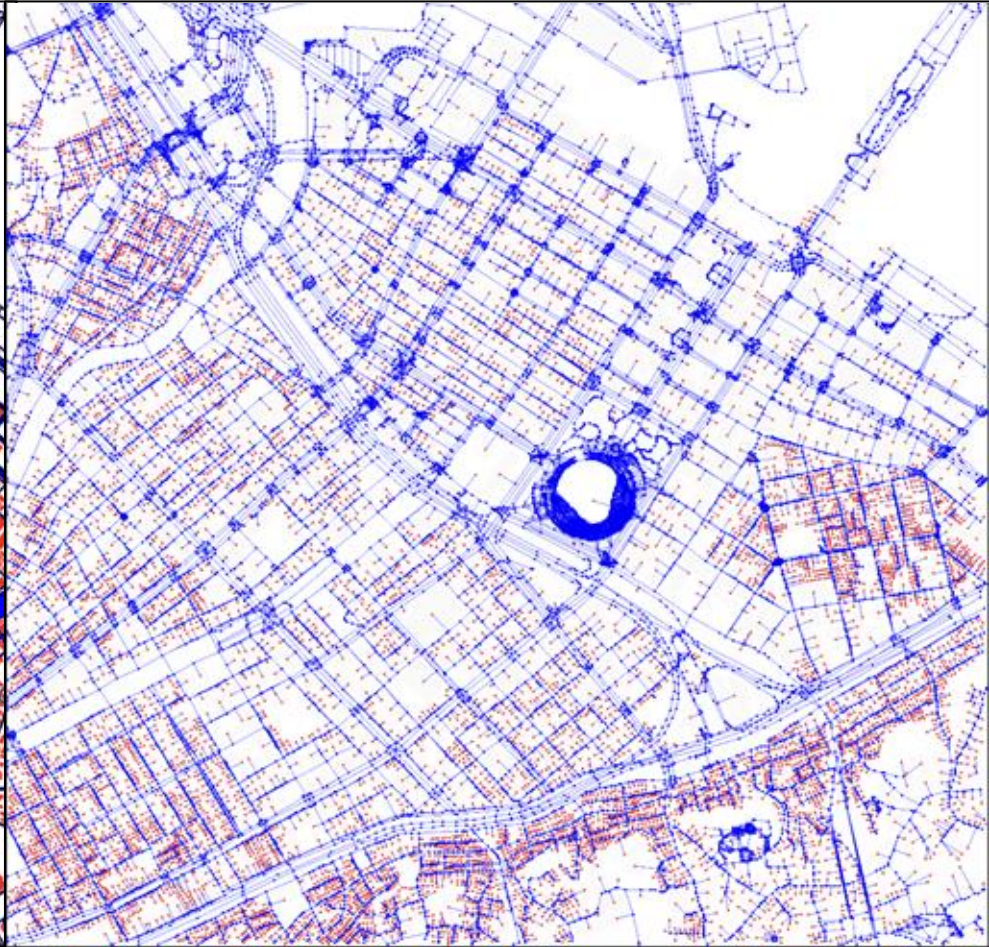
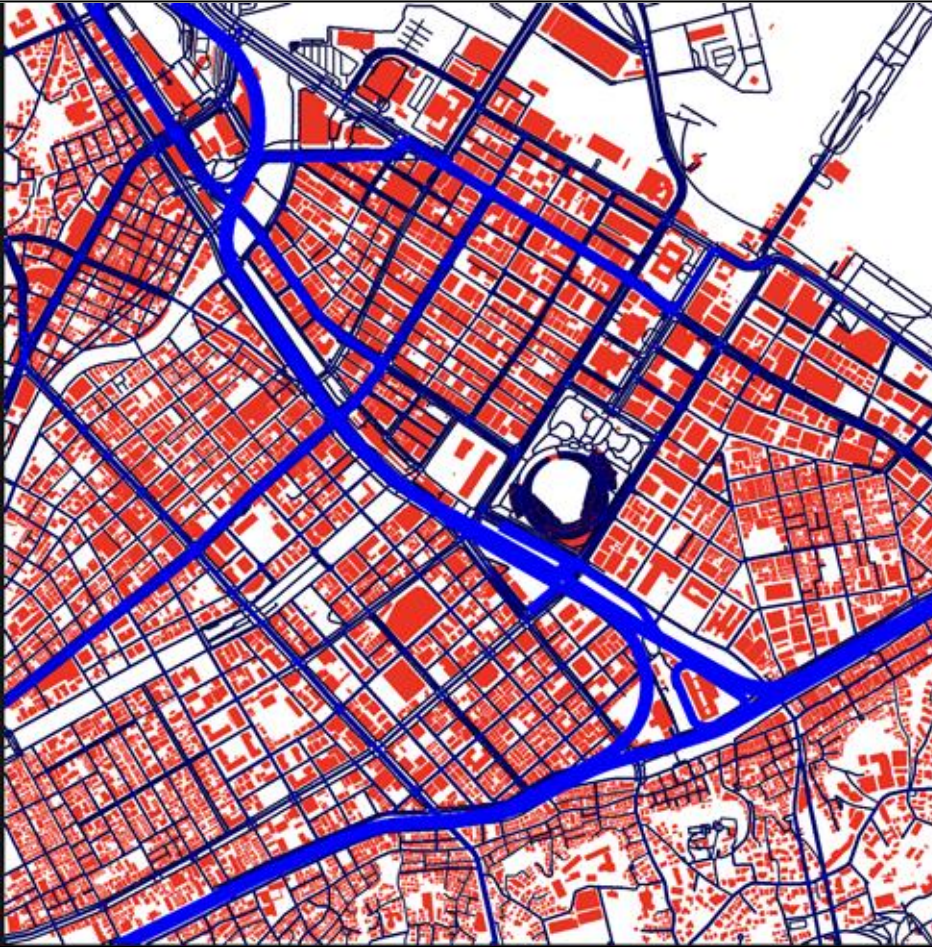
- **Not a design principle itself — a measurement space for describing and comparing morphological difference**
- **Enjoy pressing right up against the phenomenon**
 - Do not shy away from data volume and complexity
 - Interpretation and refinement after clustering matter most
 - Data and sensibility no LLM can take

takeaway-2

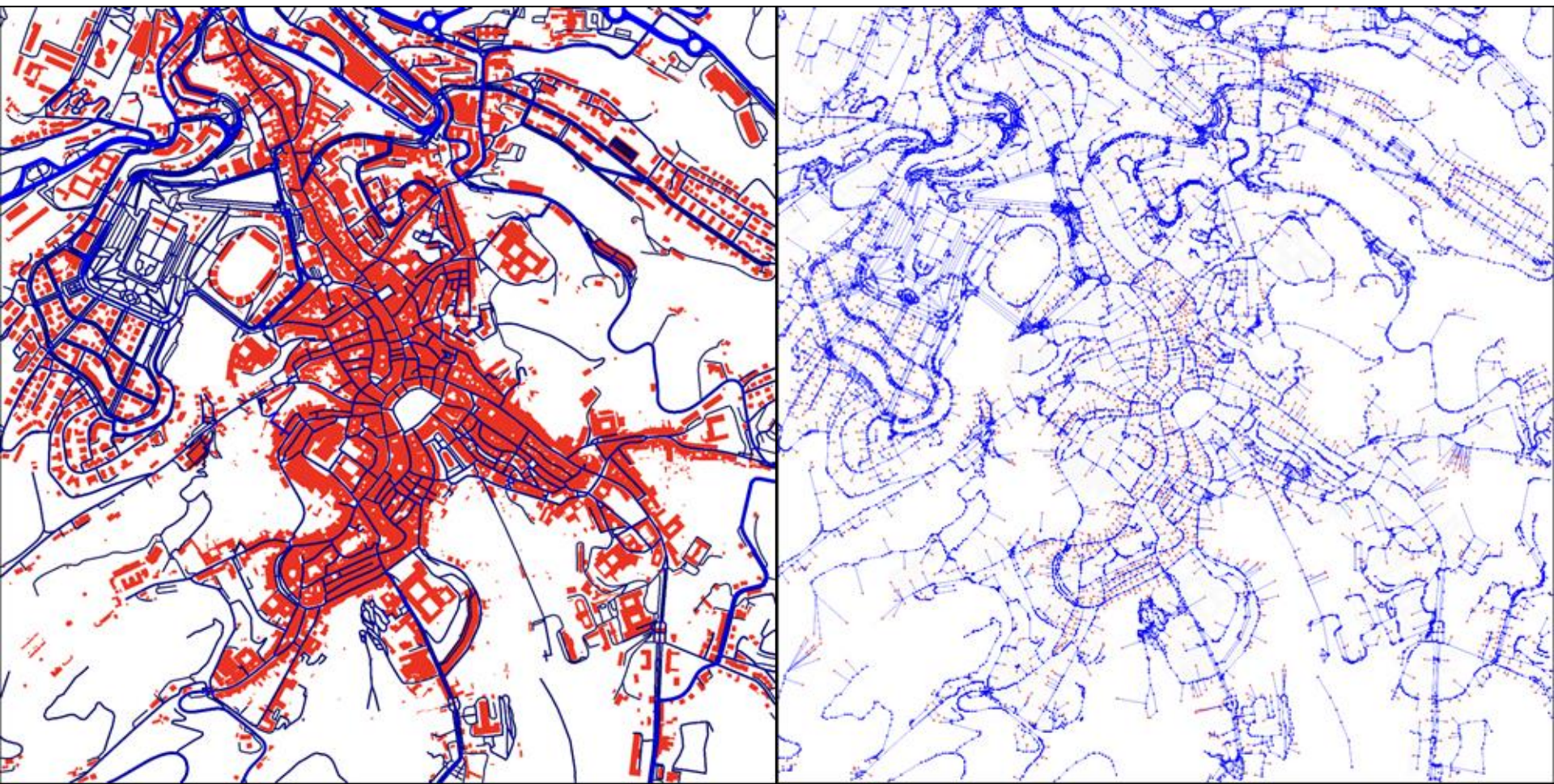
- **Diverge, then converge**
 - Ideas keep expanding; context and implementation get messy — tidy up weekly
- The issue matters, of course — but data and method matter just as much
 - OpenStreetMap → Overture
 - Cross-field methods: life and cities, metamaterials

appendix

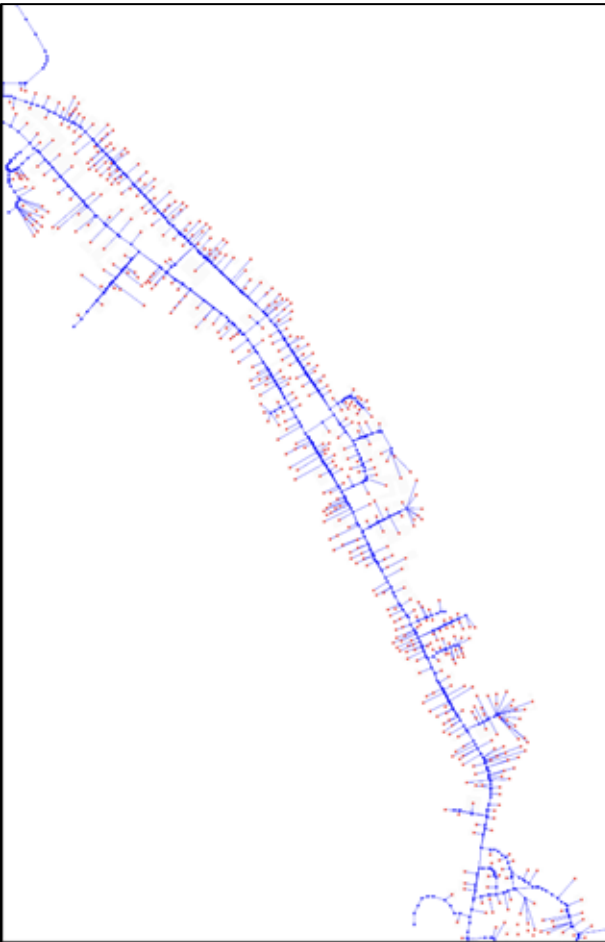
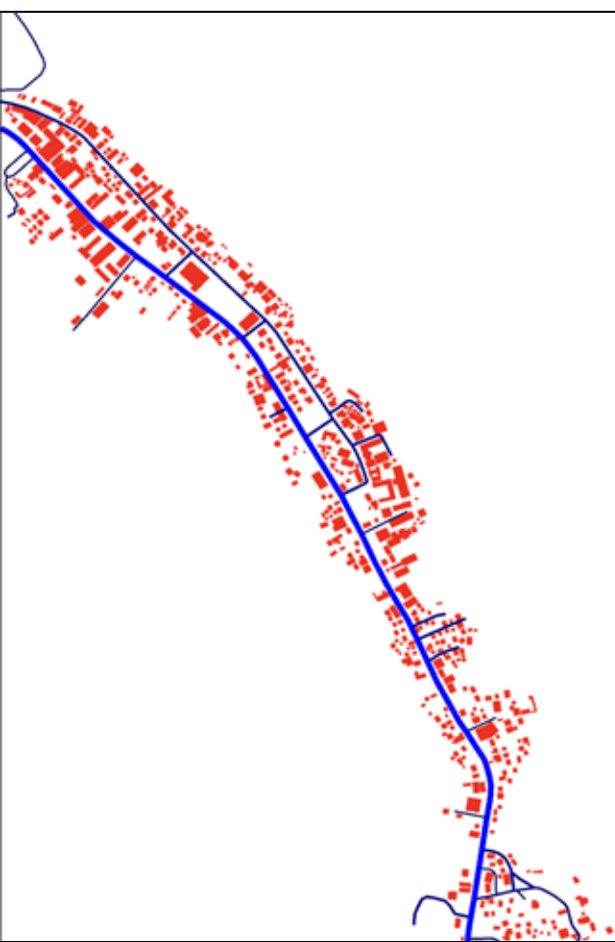
Yokohama



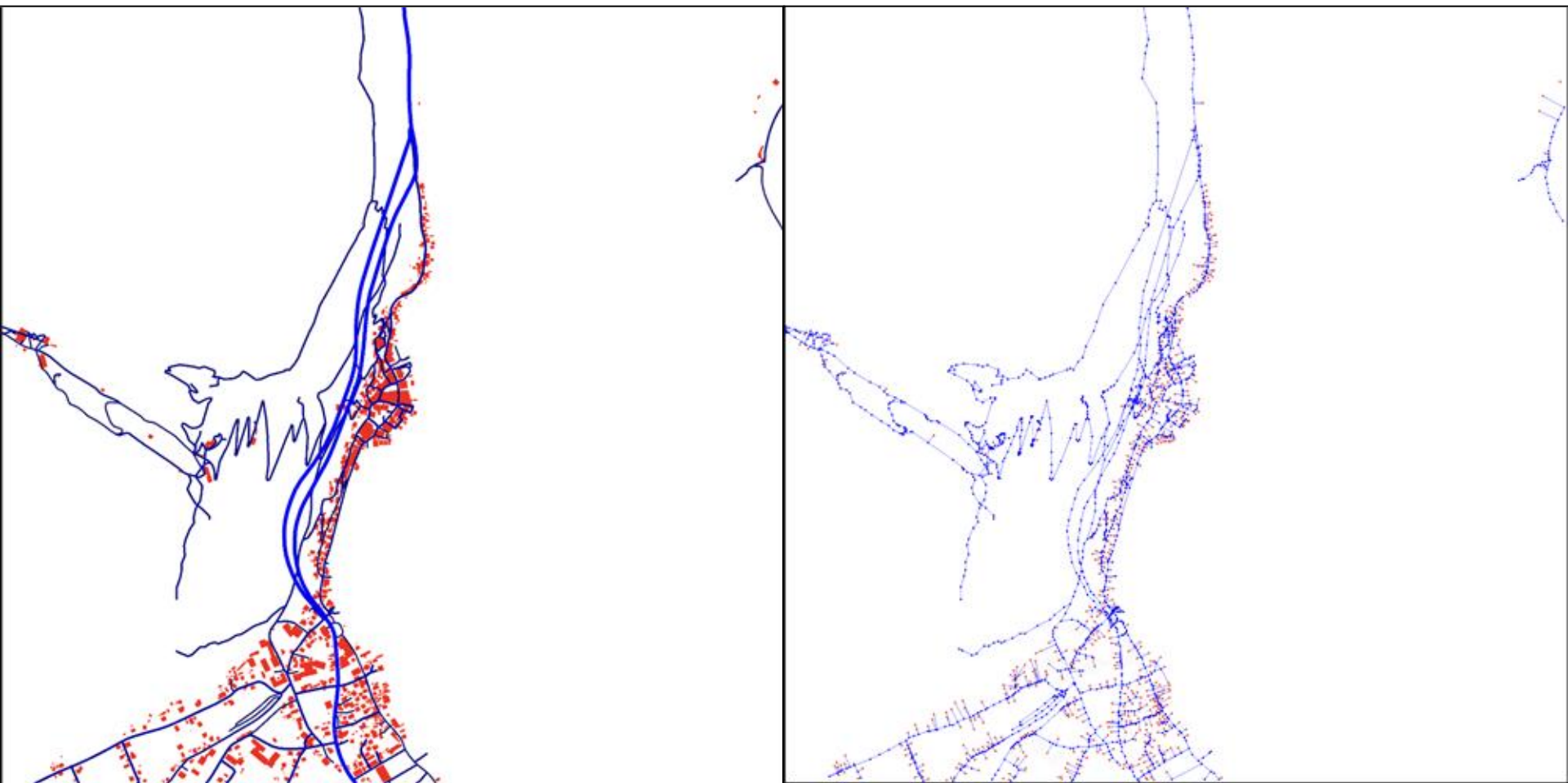
Siena



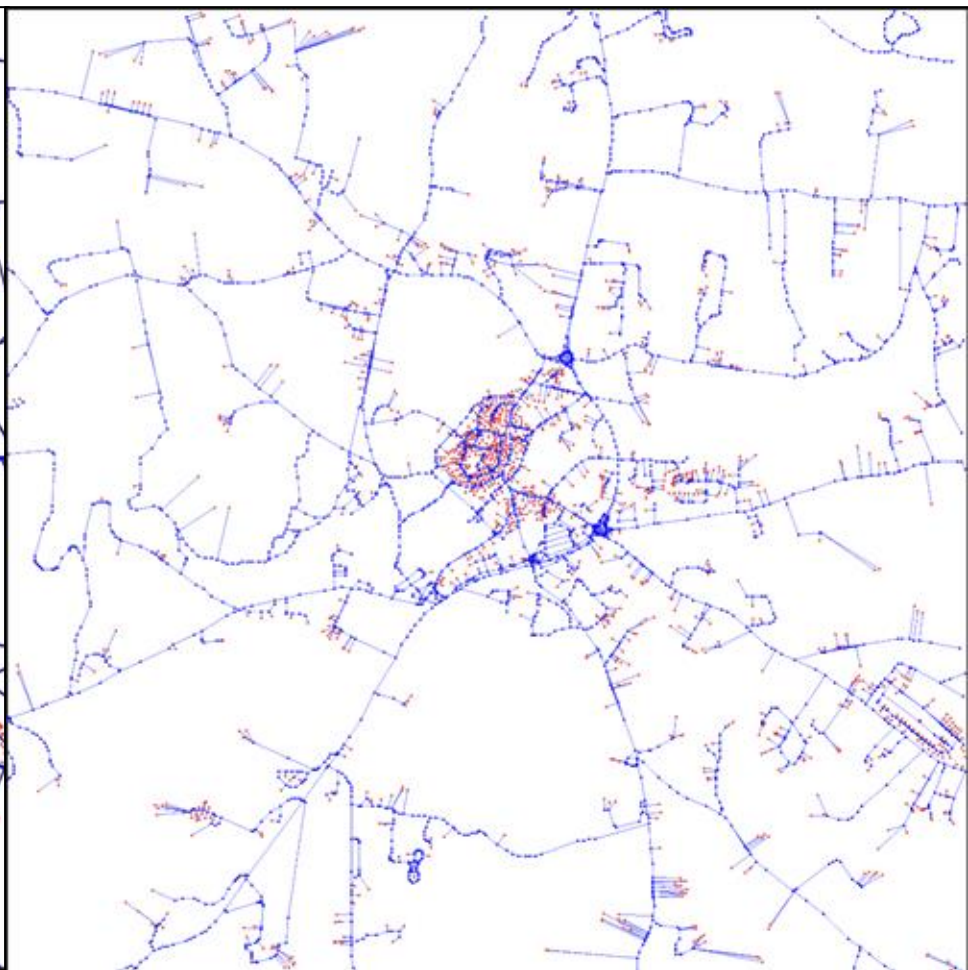
Majuro



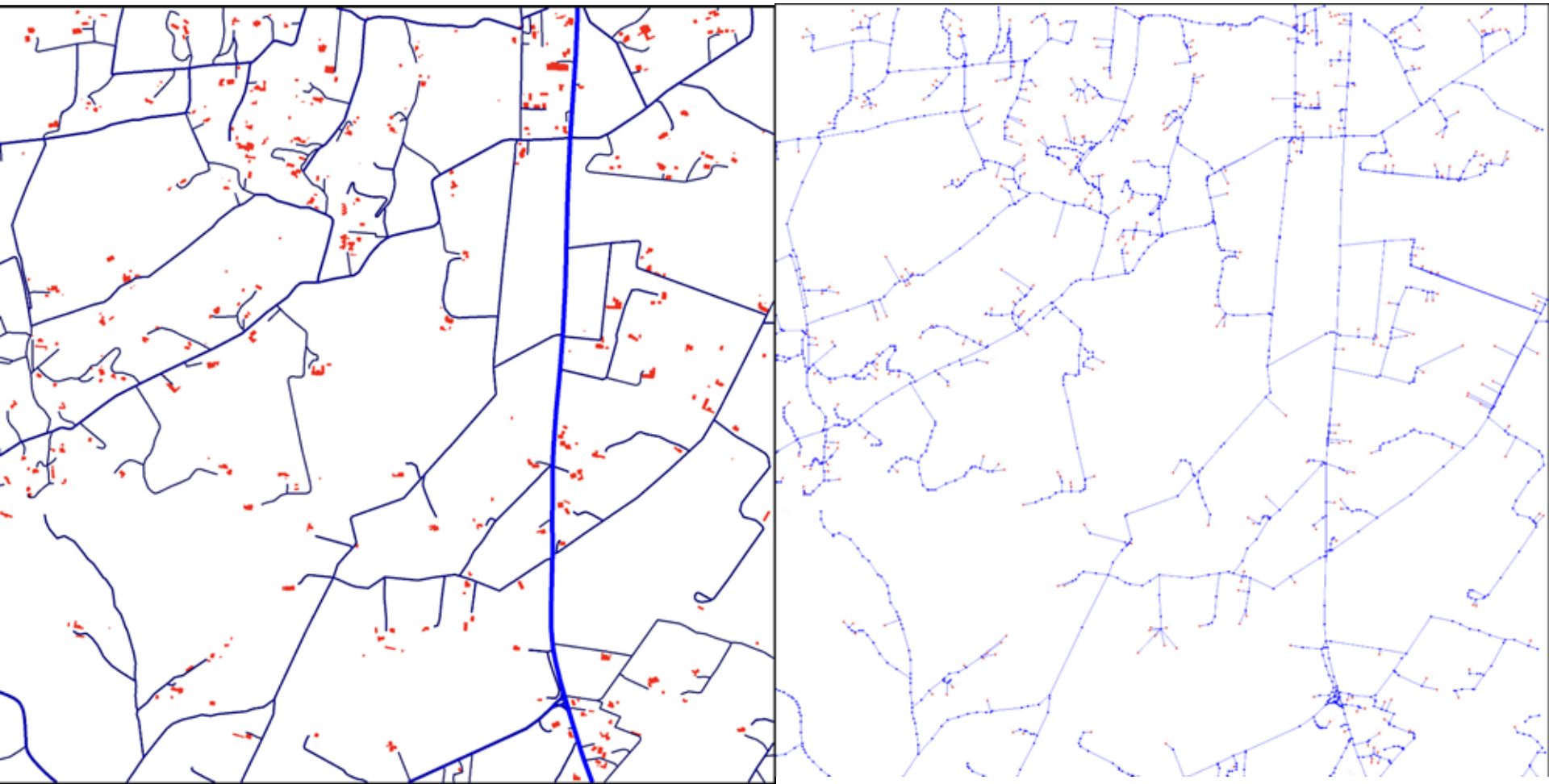
Hallstatt



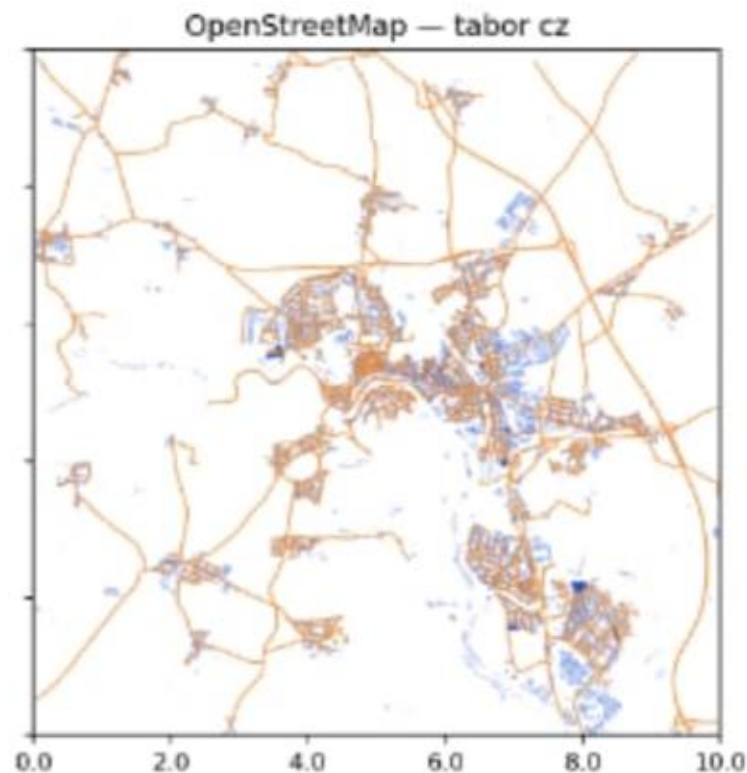
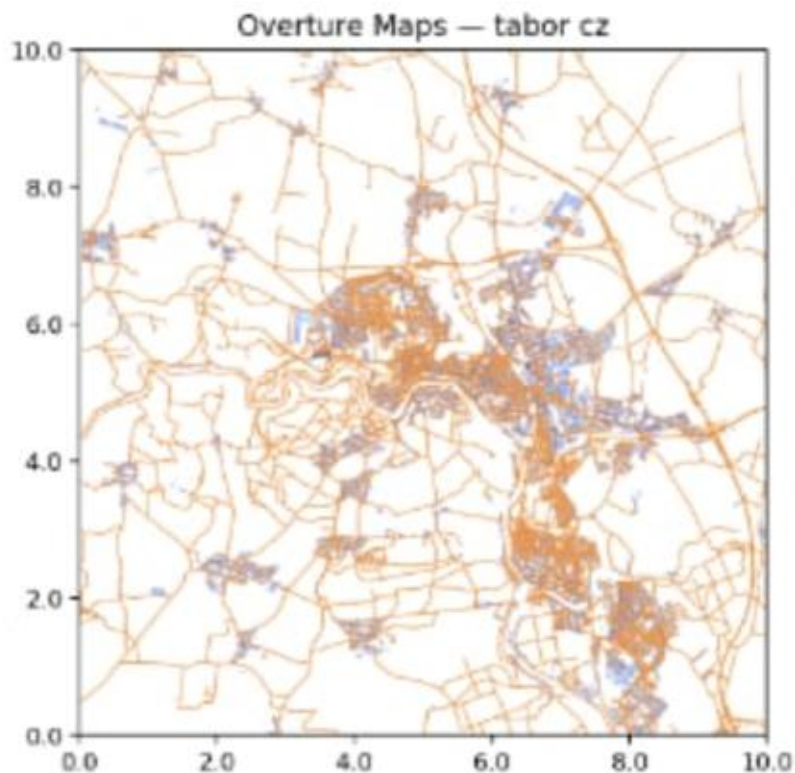
fourmarin



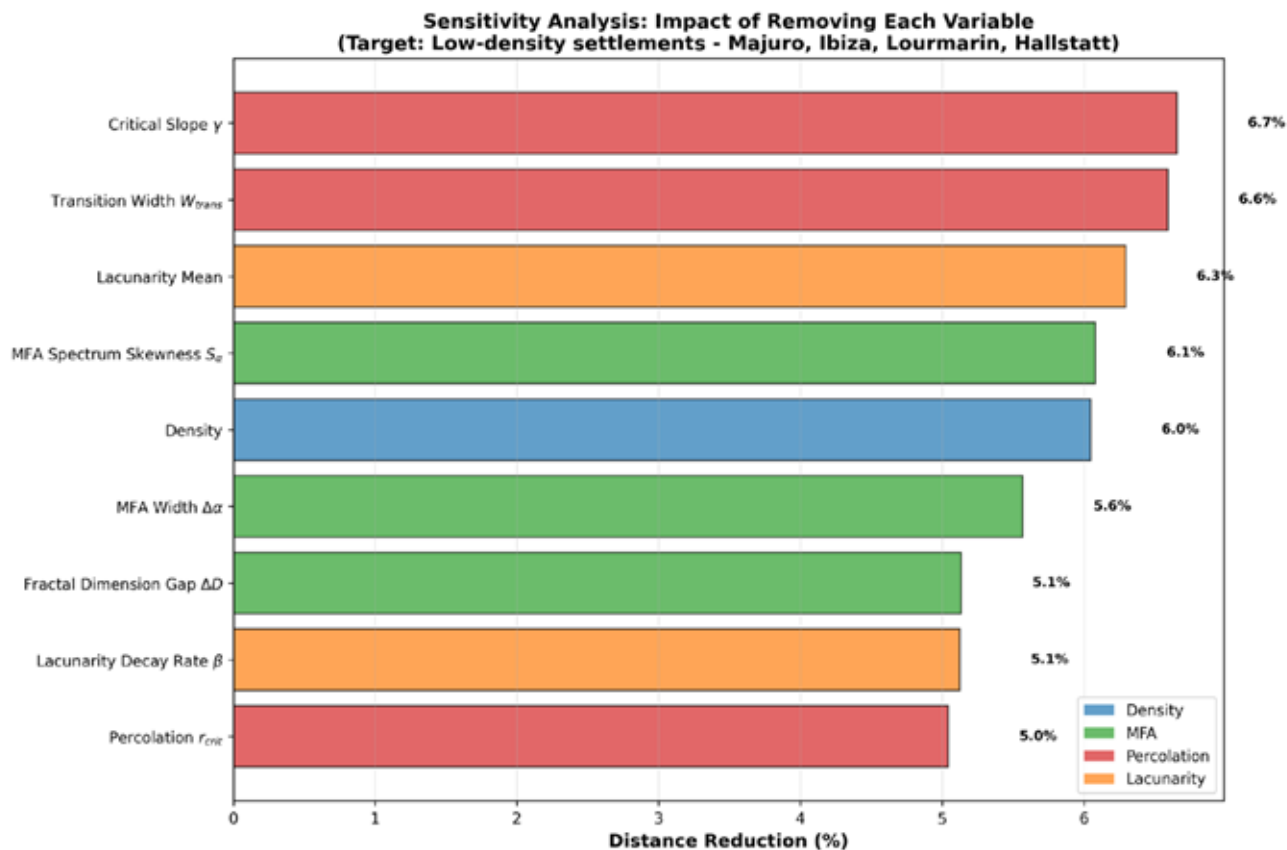
Ibiza



Richness of road & building data: Overture > OpenStreetMap



6-site clustering sensitivity: indicators much or beat density



Inspiration from metamaterials

